HIKET — data corrections and ablation tests

Working document: what was tested, what it showed, what was decided

August 2026 — working document, not for circulation

Abstract

Record of the pre-recalibration work of 4–5 August 2026. Two upstream data errors were found and corrected — a missing coarse-fragment correction in the SOC stocks, and an artefactual first year in the litter series — and a set of ablation experiments was run to attribute the effects of the round-1 changes (C1–C5) before committing cluster time. The headline consequence of the data work is that the observed 1985→2006 rise in soil carbon falls from +61% to +12%. The headline consequence of the ablations is that C5, the down-weighting of the 1985 campaign, changes predictive skill by under 1% while moving σ_{init} by a factor of about four, and is therefore dropped. Eight plots whose litter series is a single repeated value are also excluded, leaving 512 calibration-ready plots.

Contents

1 Two upstream data errors

Both were present in every calibration since April 2026, including the production posteriors `20260710_*`. Both were invisible to the existing checks. Full methods text lives in the METHODS & MATERIALS block of `Data/Data_work.R` and in the manuscript; only the essentials are repeated here.

1.1 SOC stocks: missing coarse-fragment correction

The mineral layers carried no stoniness correction, over-counting mineral carbon by a near-constant factor of 1.60–1.66 across all three mineral layers in both years, against an organic-layer ratio of 0.86. With a median coarse-fragment content of 43%, $1/(1-0.43) = 1.76$ accounts for the gap. Separately, the three campaigns had drifted onto different processing paths — the 2024 ingest was coded against the *1985* layer protocol.

The fix was to take all three campaigns from one LUKE source (H. Ilvesniemi's workbook), which reproduces the official national figures exactly (59.1 and 61.0 $\mathrm{Mg\,ha^{-1}}$ for organic + 0–40 cm, region-weighted, 2006 and 2024, $n = 446$). The calibration target is a whole-profile stock capped at the depth augering actually reached, rather than extrapolated to a fixed 1 m; the deep tail is validated against the *measured* 40–80 cm BioSoil layer (held out, 501 plots, median pred/obs 0.96).

Table 1: Effect on the calibration target (campaign medians, $\mathrm{tC\,ha^{-1}}$).

	1985	2006	2024
superseded target	63.3	102.0	105.3
homogenized target	**59.4**	**66.7**	**67.4**
change	−6%	−35%	−36%

Calibration-ready plots rise 447 → 520 (and then to 512 after the constant-litter exclusion below); the observation CV used for the fixed likelihood σ_{obs} falls 0.472 → 0.440.

1.2 Litter: an artefactual first year

The first year of the Tupek litter product is not usable. Source median total litter is 0.060 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ in 1985 against 1.398 in 1986 — a factor of 23 — with an *identical* record count (2805 site-years) in both, uniformly across months, plots and all three size classes. No ecological process produces a single year at 4% of the next and then recovers.

This mattered disproportionately because 1985 is t_0, so one bad year contaminated four quantities at once: the pre-run *endpoint* J_{t_0} (a fifth of its five-year window), the 1917 anchor J_{full}, the flux bridge J_{bar} that converts the physical litter window into the σ_{input} multiplier, and the forward run's own first year — which lands exactly on the VMI8 observation. Uncorrected, J_{t_0} was 1.543 instead of 1.863: the spin-up terminated $\approx 21\%$ below the true 1985 litter level, biasing the initial carbon state *low* in all six models.

The fix is a per-plot, per-component linear backcast of the 1986–1990 trend to 1985, giving 1.754 against 1986's 1.838. A flat multi-year mean was rejected because litter rises across that window, so averaging would place 1985 *above* 1986 and contradict the growing-stock history used elsewhere in the initialization.

> **Unconfirmed mechanism.** The collapse is consistent with litter estimated from between-inventory biomass increments, whose first year has no predecessor to difference against, but this has *not* been confirmed with the dataset author. The treatment does not depend on the mechanism — the value is unusable either way — but the wording should be checked with B. Tupek before it reaches coauthors.

1.3 Consequence for the argument

The two corrections together flatten the observed trajectory. The apparent 1985→2006 sink falls from +61% to +12%.

This cuts both ways and should be stated as such. The data are *sounder*: a +61% jump in two decades was never physical, and the QC now explains it as cross-campaign inconsistency rather than soil carbon. But the effect the paper is built on is *smaller* — roughly a fifth of what the old figures implied. The argument survives, because a steady-state start predicts a flat trajectory and the data show +12% then flat, but the rhetoric must be recalibrated to the true effect size.

2 Test 1 — C3 (non-linear historical inputs), forward-only

2.1 Design

C3 replaces the linear 1917→1985 litter ramp with a shape derived from the NFI growing-stock reconstruction. Because it changes the spin-up *forcing*, its effect is visible with all parameters *held fixed* — no MCMC needed. The engine's own forward path was run twice per plot at identical parameters, once with the growing-stock shape and once with it stripped so the wrappers fall back to linear.

A faithfulness guard reconstructs the total log-likelihood from the forward pass and refuses to report unless it matches the engine's `ll_fn` exactly. It matched to 0 — after catching two errors in the harness (below).

2.2 Result: C3 is second-order

The effect on the initial state, by σ_{init}, across all six models:

Table 2: C3 effect on `C_init` ($\mathrm{tC\,ha^{-1}}$): growing-stock shape minus linear ramp.

σ_{init}	SP1	TP2	TP3	Yasso07	Yasso15	Yasso20
0.20	−9.9	−3.4	−3.4	−7.3	−6.4	−3.5
0.30	−7.1	−2.3	−2.3	−6.0	−5.5	−2.7
0.50	−2.5	−0.7	−0.7	−1.2	−1.7	−1.4
0.70	+1.8	+0.5	+0.5	+2.9	+2.6	+1.1
1.00	+9.5	+2.7	+2.7	+7.4	+7.5	+3.8

Three findings, consistent across all six models. C3 moves the initial state by only 2–10 $\mathrm{tC\,ha^{-1}}$, whereas σ_{init} moves the same quantity by 40–80 over its plausible range — an order of magnitude more leverage. C3's effect *changes sign* at $\sigma_{\text{init}} \approx 0.6$–$0.7$, the same crossover in every model. And TP2 and TP3 are numerically near-identical at the ICBM anchor, which is C1 working as designed (TP3's S+H subsystem together represent ICBM's "old" pool), not a bug.

A framing error, corrected. This was initially written up as "C3 fails to explain the +26 $\mathrm{tC\,ha^{-1}}$ over-prediction". That misstates C3's purpose. `REVISION_PLAN.md` justifies C3 as *the physically honest pre-observation forcing*, with the gap test as a hoped-for consequence. The linear ramp was indefensible regardless of what replacing it does to the gap. The correct reading is milder and more useful: C3 is conceptually right *and* quantitatively second-order, so it neither needs defending on fit grounds nor much confounds the 1985 attribution.

2.3 A structural point that emerged

σ_{init} is not merely "uncertainty on the initial state". Because $J_{1917} = J_{\text{full}} \cdot \sigma_{\text{init}} \cdot \sigma_{\text{input}}$ while $J_{1985} = J_{t_0} \cdot \sigma_{\text{input}}$, and $J_{\text{full}} > J_{t_0}$ (litter rises after 1985), σ_{init} silently controls whether the reconstructed 1917→1985 history *rises or falls*. The threshold is $\sigma_{\text{init}} \approx 0.826$ on the median plot; above it the pre-run declines, contradicting the growing-stock history C3 encodes. The threshold is plot-specific (IQR 0.57–1.00) while σ_{init} is global, so no single value satisfies all plots. Worth a sentence in the methods regardless of what else is decided.

3 Test 2 — the ablation suite

3.1 Design and what is comparable

Six configurations per model, each differing from the reference in exactly one factor: `A0` reference; `A1` C5 off; `A2` C5=1.5; `A3` C5=3.0; `A4` C3 off; `A5` the 1985 campaign withheld from the likelihood entirely (LOCO). Control is by environment variable, with defaults bit-identical to production when unset.

Log-likelihoods are not comparable across C5 settings. C5 changes the observation SD and therefore the normalising constant of the density itself, so a config with inflated SD gets a mechanically different ℓ regardless of fit. Nothing read off the likelihood (DIC, WAIC, chain ℓ spread) can rank these configurations. Judgement rests on *unweighted* per-campaign RMSE and bias computed after the fact.
Two R^2 definitions are in play. `run_*_predictive.R` reports $\mathrm{cor(obs, pred)}^2$, which ignores bias entirely — that is the source of the documented "R^2 0.05–0.11". Variance-explained R^2 is far harsher and goes negative here because of the systematic $\approx +10$ $\mathrm{tC\,ha^{-1}}$ over-prediction. Both are now reported. On the production metric TP2's reference scores 0.023: low, but the same order as before, not a collapse.

3.2 Convergence: what is usable

Table 3: Convergence at 3 chains × 6000 iterations.

model	max $\hat{R}$	status
TP2	1.025–1.081	**clean — all six configs usable**
SP1	1.051–1.376	usable except `A0`/`A4` ($\hat{R}$ 1.34/1.38)
Yasso07	1.399–2.314	**not converged — excluded**
Yasso20	1.524–2.225	**not converged — excluded**

The setting was calibrated on TP2's seven free parameters and is inadequate for the Yasso models' ≈ 19. Their configurations `A0`, `A1` and `A5` are being re-run at 20 000 iterations; the 6000-iteration results are not reported anywhere.

A process error worth recording. TP2's `A1` was lost because the run scripts were edited *while the suite was launching processes from them*; it read a truncated file and never saved a posterior. `A2` crashed after saving and its posterior is valid. Both were re-checked, and `A1` was re-run. Rule: never edit run scripts while a suite is live.

4 Results — C5

4.1 The complete sweep (TP2, all configs converged)

Table 4: Unweighted fit at each configuration's posterior median. "Trusted" is the 2006+2024 campaigns; "1985 excess" is the 1985 bias minus the model's bias on the campaigns it did see.

configuration	trusted RMSE	1985 excess	σ_{init}	σ_{input}
`A1` C5 off	44.83	+3.19	—	—
`A2` C5 = 1.5	45.03	−1.68	0.381	1.633
`A0` C5 = 2.0	44.63	−5.43	0.285	1.887
`A3` C5 = 3.0	45.26	−10.29	0.199	2.211
`A5` LOCO	45.75	+6.32	0.780	1.224
`A4` C3 off	43.26	+3.00	0.469	1.548

C5 does not affect predictive fit *of the stock.* Turning it off entirely changes trusted-campaign RMSE from 44.63 to 44.83: **0.45%**, against a $\approx 2\%$ noise floor at this chain length. The full range spans 1.4%. SP1 independently gives a spread of **0.25%** (47.35–47.47) across the same settings.

Corrected 2026-08-07: this holds for RMSE on the stock and fails for the stock ***change****, which C5 controls strongly enough to flip its sign. See the addendum at the end of this document.*

There is no 1985 anomaly to correct. The bias-controlled excess is small from both directions: forced to take 1985 fully seriously (C5 off) the model over-predicts it by +3.2 relative to its general bias; never having seen it (LOCO) by +6.3. Both are below the +8–10 threshold set *before* the results were seen. SP1's LOCO gives +13.0, which would support C5 — but from $\sigma_{\text{init}} = 1.126$, i.e. a 1917 flux exceeding the whole-record mean, past the inversion threshold and contradicting the growing-stock history. That answer comes from an unphysical corner and is read as SP1's single pool straining without the 1985 anchor.

But C5 substantially moves the reported parameters. σ_{init} ranges 0.199 to 0.780 across the C5 settings — nearly a factor of four — and σ_{input} from 2.211 to 1.224. These are the paper's headline quantities.

The bias control is a judgement, made explicit. The pre-agreed rule was "LOCO predicts 1985 above observed by $\geq$ 8–10". Raw, LOCO gives +18.85, which clears it. But the same configuration over-predicts 2006 by +14.13 and 2024 by +10.75 — campaigns it *did* see. Without controlling for that general bias, any positively-biased model would "prove" that VMI8 reads low. The controlled excess is +6.3. The adjustment changes the answer, so it is recorded rather than absorbed silently.

4.2 Confirmation on a third model (Yasso07, 20 000 iterations)

The 6000-iteration Yasso runs did not converge ($\hat{R}$ up to 2.31) and were discarded. Re-run at 20 000 iterations they are clean ($\hat{R}$ 1.002–1.034 for `A0`, 1.001–1.106 for `A1`), and they reproduce the result:

Table 5: Yasso07, C5 on vs off. Fitted on the 520-plot bundle (before the constant-litter exclusion); both configurations are scored identically, so the contrast is internally valid.

configuration	trusted RMSE	1985 excess	σ_{init}	σ_{input}
`A0` C5 = 2.0	46.39	−5.31	0.440	1.320
`A1` C5 off	46.25	+4.98	0.767	0.945

Turning C5 off changes trusted-campaign RMSE by 0.30%. Across the three models that converged the effect is 0.45% (TP2), 0.25% (SP1) and 0.30% (Yasso07) — all far inside the noise floor, in a two-pool model, a single-pool model and a Yasso structure that parks carbon in slow humus. The bias-controlled 1985 excess with C5 off is likewise small everywhere: +3.19, +2.92 and +4.98. And σ_{init} moves substantially in every case (0.440 → 0.767 here), confirming that C5's real effect is on the reported parameters rather than on prediction.

Yasso07's σ_{input} falls below 1 when C5 is removed (1.320 → 0.945), joining SP1 (0.23–0.61) on the wrong side of the C4b prior, which is centred at 1.30 on the argument that understorey litter is *missing*. With C5 off, σ_{init} also reaches 0.767, close to the 0.826 pre-run inversion threshold. Neither blocks the C5 decision, but both belong in the discussion of what σ_{input} is really absorbing.

4.3 Why C5 no longer stands up

Its supports fail one by one. The original justification — that VMI8 mineral stocks read systematically low — is now *contradicted*: the SOC homogenization removed most of the 1985→2006 jump, and neither endpoint of the sweep finds an anomaly at 1985. The factor 2.0 was grounded in the ≈ 30% discrepancy that the stoniness fix removed, so the number has no remaining basis. And the mechanism does not match the claim: inflating *independent* per-observation σ mostly averages away over 441 plots (the campaign mean's SE scales as $\sigma/\sqrt{n}$), whereas a protocol difference produces a *shared* offset whose uncertainty does not shrink with n at all.

What remains is a legitimate concern of a different kind: 2006 and 2024 were validated against official LUKE figures and 1985 could not be, and the protocols genuinely differ (VMI8 samples 0–5 and 5–20 cm; the later campaigns 0–10 and 10–20 cm). That is a statement about confidence in the data. It is real, and it belongs in the limitations — not in a tuning constant that reshapes σ_{init} without improving prediction.

DECISION — C5 is dropped

The 1985 campaign is **not** down-weighted. `SIGMA_1985_INFL` is set to 1.0 and C5 is removed from the pipeline before the Roihu recalibration.

Rationale. C5 changes predictive skill by under 1% while moving σ_{init} by nearly a factor of four. A knob that does not improve prediction but substantially moves the reported parameters is a researcher degree of freedom. The correction it was built to make has already been made *in the data*, which is where it belongs.

What replaces it. The sensitivity is reported rather than a value chosen: down-weighting the VMI8 campaign changes predictive skill by under 1% but shifts σ_{init} from 0.29 to 0.78 across the plausible range. Inferences about the initial state are correspondingly sensitive to confidence in the 1985 campaign. This is strictly more informative than picking a factor, and it costs nothing now that the sweep exists.

5 Side findings

5.1 C3 costs a little fit

`A4` (C3 off) gives the best trusted-campaign RMSE in both usable models: TP2 43.26 vs 44.63 (3.2%), SP1 44.72 vs 47.43 (6.1%). Two qualifications matter. Both SP1 configurations in that comparison are among the ones that did not converge ($\hat{R}$ 1.34/1.38), so only TP2's 3.2% is trustworthy. And the configurations differ in more than the pre-run shape — the calibration re-optimised around it (σ_{init} 0.285 → 0.469, σ_{input} 1.887 → 1.548) — so most of the gap is a bias reduction attributable to that shift, not to C3 in isolation.

The clean attribution remains the forward-only test, where C3 moved `C_init` by 2–10 tC ha^{-1} at fixed parameters. A ≈ 3% RMSE effect, partly absorbed by recalibration, is consistent with that. It is a footnote for the robustness appendix, not a reason to reconsider C3.

> **DECISION — C3 is kept**
> C3 stays. It is justified by the growing-stock record, not by fit; the linear ramp was indefensible regardless. The ≈ 3% fit cost is reported honestly rather than omitted.

5.2 σ_{input} diverges sharply by structure

SP1 lands at 0.23–0.61 — tree litter roughly twice too *much* — while TP2 lands at 1.2–2.2, too little. SP1 is therefore in direct conflict with the C4b prior centred at 1.30 on the argument that understorey litter is *missing*. This sharpens the existing σ_{input} result rather than contradicting it, but the conflict should be addressed rather than noted in passing.

5.3 Input QC sweep

A repeatable sweep (`doublechecks/input_data_qc.R`) was written on the principle that the 1985 artefact survived every existing check because none was looking at the *ends* of series, year-on-year steps, or per-plot constancy. Results: no negative or missing litter, no duplicate rows, complete coverage, sane climate, and no anomalous year-steps.

Two findings. First, 37 plots with near-zero litter through 1985–1995 are *real* — young or recently clearcut stands, 36 of 37 recovering more than tenfold. These were initially flagged as a serious problem on the grounds that J_{t_0} collapses for them; that reasoning was wrong, applying a steady-state argument to a transient pre-run in which humus retains carbon from earlier years. Tested directly, these plots fit 1985 *better* than the rest (−1.7 vs +16.9 tC ha^{-1} bias). A genuine point survives: the 1985 residual correlates with J_{t_0} ($r = 0.47$), so the modelled initial state is anchored on *recent* litter and recently-disturbed plots carry no memory of the previous rotation. Here that offsets the general over-prediction by coincidence, not correctness.

Second, and **unresolved**: ten calibration-ready plots have *literally constant* litter for 39 years, identical to four decimal places, constant in the source as well. No stand produces identical litter for 39 consecutive years; the litter model most likely returned a fixed value where the underlying biomass series was static. They inject a spurious "no temporal change" forcing into a study whose subject is the temporal change. Plots: 19571, 35391, 37591, 39251, 41351, 43651, 43771, 45331, 61591, 67631. To be raised with B. Tupek alongside the 1985 question; flag-and-keep versus exclude is not yet decided.

6 The delta-offset test

C5's mechanism never matched its claim: inflating *independent* per-observation σ averages away over 441 plots, whereas a protocol difference is a *shared* offset whose uncertainty does not shrink with n. The faithful form is a campaign-level offset, $\text{obs}_{1985} \sim \mathcal{N}\Big((1+\delta)\widehat{\text{SOC}},\, (1+\delta)\widehat{\text{SOC}}\,\sigma_{\text{obs}}\Big)$, with $\delta < 0$ meaning VMI8 reads low. `REVISION_PLAN` had ruled this out as non-identifiable against σ_{init}. It was tested with C5 off and a deliberately wide prior, $\delta \sim \mathcal{N}(0, 0.25)$; the likelihood was validated against the engine's own to $|\Delta| = 0$ at $\delta = 0$ before sampling.

δ *is* identifiable — the revision plan was too strong. Posterior $\delta = -0.137\ [-0.197, -0.080]$, sd 0.030 against a prior sd of 0.25 (ratio 0.12), $\hat{R} = 1.004$, $P(\delta < 0) = 1.000$.

But it is not trustworthy as a measurement offset. Three things disqualify it. The trusted campaigns are *identical* with and without δ (RMSE 44.18, bias +9.30 either way), because δ touches only 1985 — so it is informed by the 1985 residual alone, and improving that residual by adding a free 1985 parameter is circular. It buys its fit with an unphysical history: σ_{init} rises to 0.98, with 80% of draws past the pre-run inversion threshold, i.e. a reconstructed 1917–1985 litter trajectory that *declines*. And $\text{cor}(\delta, \sigma_{\text{init}}) = -0.64$ confirms the trade-off the revision plan anticipated.

Taken at face value it would erase the phenomenon. $\delta = -0.137$ implies a true 1985 stock of 68.8 tC ha^{-1}, making the trajectory $68.8 \to 66.7 \to 67.4$: declining, then flat — no accumulation at all. That is a far larger claim than C5 ever made, and it contradicts the homogenized data validated exactly against the official LUKE figures (51.8 in 1985 vs 58.8 in 2006, measured 0–40 cm).

> **DECISION — δ is not adopted either**
> Both C5 and δ are mechanisms for distrusting 1985. The faithful one, given freedom, returns an implausible answer and pays for it with a history we know to be wrong. A bias parameter informed only by the residual it explains is a misfit sink. Confidence in the 1985 campaign is handled in the limitations.
>
> *Worth reporting regardless:* if the model is allowed to treat 1985 as offset, it concludes there was no accumulation. The accumulation signal depends on trusting that campaign — a striking sensitivity, and an honest one to state.

7 Constant-litter plots

The input QC sweep found plots whose modelled litter is a single value repeated unchanged for 1986–2024 (relative SD $< 10^{-6}$), constant in the source as well. No stand produces identical litter for 39 consecutive years; this is the litter model returning a fixed value where the underlying biomass record was static. It is the same class of defect as the artefactual first year — plausible, positive, finite and wrong — and it survived every earlier check because nothing tested per-plot constancy.

Sixty-four such plots exist dataset-wide; eight were calibration-ready.

> **DECISION — Eight constant-litter plots excluded**
> Excluded rather than flagged: a constant series injects a spurious "no temporal change" forcing into a study whose subject *is* the temporal change. Cost: 8 of 520 plots. Calibration-ready falls $520 \to 512$. Detected by rule rather than hard-coded, so it survives a change in the litter product.
>
> *Borderline, deliberately kept:* plots 39251 and 67631 take only three distinct litter values across 39 years (relative SD ~ 4–7×10^{-3}). Suspiciously quantised but not constant, so outside the rule. Recorded so the boundary is visible rather than implicit.

8 What remains open

- **Yasso07 and Yasso20 ablations** re-running at 20 000 iterations. They are confirmatory now that C5 is decided, but they test structurally different models — Yasso parks carbon in slow humus — so a contrary result would be worth knowing.
- **The δ-offset test.** The faithful representation of a protocol difference is a campaign-level offset $\text{obs}_{1985} = (1+\delta)\,\widehat{\text{SOC}}$ rather than inflated independent noise. The revision plan ruled this out as non-identifiable against σ_{init}; that may be too strong, since σ_{init} affects all three campaigns while δ affects only 1985. Cheap to test and would settle whether the concern can be expressed faithfully at all.
- **C1 remains unattributed.** The largest round-1 change, and its headline claim — that anchoring rates externally makes σ_{input} fall *as a consequence* — is untested.
- **The measured-only (0–40 cm) target** would retire the depth-extrapolation question entirely for the robustness appendix. Cheap: a column swap.
- **Confirmation from B. Tupek** on the 1985 first-year artefact and the ten constant-litter plots.
- **Corroboration of the stock QC** from A. Lehtonen and J. Heikkinen. The QC itself is ours and was performed here; their role is review.

9 Addendum, 2026-08-07: C5 was adjudicated on the wrong metric

The C5 decision above stands, but one of its stated grounds does not. The claim that C5 "does not affect predictive fit" was established on trusted-campaign **RMSE**, which is a metric on the SOC *stock*. σ_{init} barely moves the stock. What it moves is the direction of the 1917→1985 pre-run, and therefore the **stock change** — the quantity a national greenhouse-gas inventory actually reports, and the one C5 turns out to control. The original sweep never measured it.

The pre-run inverts at a threshold set by the litter record, not by any model. The transient initialisation starts each plot at steady state under the 1917 flux $J_{1917} = J_{\text{full}}\,\sigma_{\text{init}}\,\sigma_{\text{input}}$ and ramps it to the 1985 flux $J_{1985} = J_{t_0}\,\sigma_{\text{input}}$. Because the pre-run begins *at* equilibrium and the flux then moves monotonically, SOC moves monotonically with it. The pre-run therefore declines exactly when

$$\sigma_{\text{init}} \;>\; J_{t_0}/J_{\text{full}},$$

with σ_{input} cancelling. The per-plot ratio has median 0.818 over the 512-plot bundle, reproducing the 0.826 quoted earlier from the ablations. Nothing in this criterion depends on the decomposition model.

Stock change crosses zero at that same threshold, in every model tested. Re-running the ablation posteriors through the genuine forward model and computing paired per-plot stock change (only plots observed in both campaigns, so composition cannot drive it):

Table 6: Paired 1985–2024 stock change, $\mathrm{t\,C\,ha^{-1}\,yr^{-1}}$, at each config's posterior-median parameters. Observed +0.309 on the same 256 paired plots.

config	SP1	TP2	Yasso07	Yasso20
`A3` C5 = 3.0	+0.638	+0.625	—	+0.441
`A0` C5 = 2.0	+0.464	+0.461	+0.423	+0.203
`A2` C5 = 1.5	+0.349	+0.333	—	+0.058
`A1` C5 off	+0.164	+0.167	+0.098	−**0.103**
`A5` no 1985	−**0.184**	+0.054	—	−**0.317**
observed +0.309				

The predicted sink is monotone decreasing in σ_{init} and crosses zero at $\sigma_{\text{init}} \approx 0.8$ in all four models — SP1 (single pool) and Yasso20 (twelve free fractions) alike — confirming that the threshold is a property of the litter data rather than of kinetic structure.

Consequence for the production run. With C5 off, the 2026-08-05 recalibration puts Yasso15 at $\sigma_{\text{init}} = 1.007$ and Yasso20 at 1.243, both past the threshold: their reconstructed pre-runs decline for 78% and 92% of plots respectively. Their forward trajectories relax downward from an overshot 1985 stock, and they report a **source** where all three campaigns show a **sink** — −0.063 and −0.155 against an observed +0.312. The sign error survives dropping 1985 entirely: on 2006–2024 alone, observed +0.209 against Yasso15 −0.183 and Yasso20 −0.220.

What this does and does not change. It does not reinstate C5. Choosing the 1985 weighting so that the stock change comes out right is circular, and the C5 value required to reproduce the observed change is not identical across models — ≈ 1.7 (SP1), ≈ 1.6 (TP2), ≈ 1.7 (Yasso07), ≈ 2.4 (Yasso20). The clustering of three of the four is close enough that it admits the opposite reading — that the 1985 campaign does warrant some down-weighting — and that reading is recorded here rather than argued away. What separates the two routes is not arbitrariness but *checkability*: a constraint on σ_{init} can be tested against independent NFI growing-stock data, whereas any C5 value can only be tested against the SOC observations being fitted.

DECISION — The σ_{init} constraint replaces the C5 sensitivity as the live question

C5 stays withdrawn. The unresolved item is not the 1985 weighting but σ_{init} itself, which is underdetermined and currently free to take values that contradict the growing-stock record the pre-run shape (C3) is built from.

Proposed. Constrain σ_{init} on the *national aggregate* ratio, $\sigma_{\text{init}} \leq \text{mean}(J_{t_0})/\text{mean}(J_{\text{full}}) \approx 0.82$, as a truncated or strongly informative prior rather than a hard cap. A per-plot form is not available: σ_{init} is a single global scalar while the ratio spans 0.009–5.983 across plots, so no global value can satisfy every plot. The bound binds only Yasso15 and Yasso20; the other four already sit inside it.

Caveats to carry into the text. (i) Constraining σ_{init} so the pre-run must rise partly *imposes* the direction subsequently reported; the defence is that the constraint comes from independent NFI data, not from the SOC observations, and this must be stated rather than implied. (ii) σ_{init} and σ_{input} are already coupled by the `flux_pair` transform, which bounds the *level* of the 1917 flux but not its *direction*; the combined feasible region should be checked before launching. (iii) A parameter pressed against a boundary mixes worse — ESS on σ_{init} needs watching, as it did for σ_{input}.

Reproducibility. `doublechecks/prerun_direction.R` reports the threshold and the share of the posterior on the declining side per model; `doublechecks/ablation_stock_change.R` reproduces the table above from the quarantined ablation posteriors; `doublechecks/production_fit_by_campaign.R` gives per-campaign fit of the production run under both R^2 definitions.